

*“The difference between theory and
practice
is larger in practice than in theory.”*

— Attributed to Jan L. A. van de Snepscheut

Abstract

The Cognitive Integrity Framework (CIF) introduced in Part 1 of this series establishes formal foundations for securing multiagent AI systems against cognitive manipulation attacks—adversarial inputs that exploit inter-agent communication to corrupt beliefs, inflate trust, or subvert coordination. Theoretical guarantees alone cannot ensure practical protection. This companion paper bridges the theory-practice gap through comprehensive computational validation: we implement the complete CIF defense suite (cognitive firewalls, belief sandboxes, identity tripwires, trust calculus with bounded delegation, and Byzantine-tolerant consensus) and evaluate detection performance across six production multiagent architectures (Claude Code, AutoGPT, CrewAI, LangGraph, MetaGPT, and Camel) using a 950-attack corpus spanning four categories with 12 subcategories. All experiments use simulation-based architecture adapters modeling topology and communication patterns of each target system; results are deterministically reproducible (seed = 42).

The layered CIF defense achieves 94% overall detection (95% CI: